

Test 1 Topics

Speaking Math — Sets — Functions

1.5-hour test

General Notes

The homework and your question compasses are a good study guide. If you understand all the homework problem, you will most likely do well.

You do **not** need to remember the exact definitions of anything. We will **not** ask you a question like “What is the definition of _____”. Just understand the intuition, be able to explain it in words, and be able to deeply logically reason about it. If needed, use the internet and AI (such as ChatGPT, Claude, Gemini, etc.) for help. You can ask something like “Give me 3 sample problems for this topic **X**,” “Explain **X** to me more, I don’t quite understand it,” “Explain **X** in more simple terms. Use an analogy,” “How does **X** relate to [an idea from a different unit],” “How did you solve this problem and why did you do this step **X**?” If you use AI, make sure to ask it **specific questions**. If you use AI, **use it as a teacher, not your answer key**.

Manage your time wisely. Communicate with Leo concerns or problems you’re facing :)

Good luck and have fun thinking.

Talking (Speaking Math)

- Expression vs. equation vs. formula
- Terms, coefficients, variables, constants, exponents, like terms
- Reading algebra aloud (“the quantity,” powers, fractions, etc.)
- Writing expressions from words

Sets

- Sets & membership ($\in$, $\notin$)
- Roster & set-builder notation
- Number sets ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$): which one are larger than the others? and why?

- Subsets ($\subseteq$): proper and improper subset
- Union, intersection, difference, complement
- Cardinality ($|A|$)
- Cartesian products ($A \times B$)
- Inclusion–exclusion
- Real-world applications

Functions

- The four representations (arrow, ordered pairs, table, graph)
- Function notation $f(x)$
- Domain, codomain, range: be able to explain the similarities and differences between all the three in words. be prepared to be asked specific questions about them.
- Injective, surjective, bijective: be able to explain the similarities and differences between all the three in words. be prepared to be asked specific questions about them.
- Proofs (injective, surjective, and bijective proof)
- Real-world applications: think of examples like the Minecraft bedroom and variations of that
- How do we compare the sizes of infinite sets?

Note: Composition, inverses, the verticle-line test, and multivariable functions will **not** be on the test.

Part 2 — New Material (Limits)

Domain & Range

- **Domain** — avoid the two dangers: no dividing by zero, no square roots of negative values.
- **Range** — set $y = f(x)$ and solve backwards for x ; which outputs are actually reachable? Understand the “story” that the equation $x = g(y)$ is telling you.
- Reading the domain and range off a graph

Limits — The Big Ideas

- What is a limit?
- One-sided limits (from the left, from the right) and two-sided limits.
- “Does not exist (DNE)” vs. “undefined”: be able to explain the difference in words, and why a limit can exist where the function is undefined (a hole)
- Reading limits from a graph: open vs. filled circles, holes, and jumps
- Finding limits from a table of values
- Finding limits with algebra: try substituting first; if you get $\frac{\text{something}}{0}$, factor and cancel
- Infinite limits and vertical asymptotes ($\pm\infty$)

Limits at Infinity (End Behavior)

- What “ $\lim_{x \rightarrow \infty} f(x)$ ” is asking, and how it shows up as a horizontal asymptote *in the long term*. (Notice we say “in the long term” because the range may still include the value of the horizontal asymptote as shown in class).
- Finding $\lim_{x \rightarrow \infty} f(x)$ with the shortcut and algebraic approach.
- Know the **limit laws**: sum, difference, constant multiple, product, quotient, power.